QVEX-NGC6503 Technical Addendum: Axial Dipole Constraint, Retrograde Vacuum Friction, and Hawking-Evaporation Density Profiles

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1 Abstract

This addendum consolidates the QVEX-NGC6503 project's axial-dipole constraint with a proposed two-front account of retrograde vacuum friction and a hypothesized radial density contribution associated with Hawking evaporation. It distinguishes the project's existing parameterization from proposed physical interpretation: the Master Note records the PID gains and friction law in §2, while the related axial node frames the dipole contribution as an extension. The retrograde friction report reports a sign transition near 12.10 kpc, with the nonlinear-friction node giving ≈ 12.11 kpc; the exact value depends on the calculation and rounding used. These elements form a coherent research architecture, but their physical coupling—especially the Hawking-to-galactic-density link and the mapping of PID gains to physical parameters—remains a hypothesis requiring derivation and independent tests.

2 Introduction and Relation to the Master Note

This addendum is intended to clarify how the project's radial and axial constraints can be discussed together without treating them as one already-validated mechanism. The Master Note, dated 2026-10-26, identifies itself as the central NGC 6503 QVEX analysis and summarizes the model parameters in §2; the document currently ends there, although related project material

points to a prospective §3. The axial-constraint node supplies the relevant dipole extension. Master Note Axial Constraint Node

The addendum therefore proposes a disciplined hierarchy: the radial component describes the fitted rotation/friction behavior, while the axial component is a separate angular constraint. Their interaction should be stated as a model extension to test, not silently incorporated into the established radial fit.

3 Methods: Formalization of the Axial Dipole Constraint

The project's working axial formulation uses a central dipole-jet term that depends on polar angle, represented in the conversation's model form as

$$V_{\text{eff}}(r,\theta) = -\frac{GM(r)}{r} + \frac{\alpha_{\text{dipole}} P_{\text{jet}}}{r^3}\cos^2\theta.$$

Here, the monopole-like radial term and the angular dipole contribution play distinct roles. Since $\cos^2(\pi/2) = 0$, the dipole term vanishes in the ideal equatorial plane; off-plane behavior depends on the sign and magnitude of its coefficient and on the full dynamical equations. Thus the expression alone does not prove stable disk confinement. That requires deriving the vertical force or restoring frequency and checking stability over the relevant parameter range.

For project organization, retain the popcorn-density and friction equations as the radial baseline, and label D_{jet} or the angular potential as an optional axial extension. The axial node should specify its assumptions, units, boundary conditions, and the observable used to test the claimed disk-plane effect. Axial Constraint Node

4 Results: Retrograde Vacuum Friction and the Transition Radius

The proposed hydrodynamic interpretation separates the interaction into a leading-front compressive contribution and a trailing-front suction contribution associated with a density deficit. In schematic form, the net radial contribution may be written as\n\n

$$F_{textnet}(r) = F_{textlead}(r) + F_{texttrail}(r),$$

\n\nwith each term's sign defined by an explicit radial convention. A change in sign occurs where $F_{textnet}(r_*) = 0$; it should be computed from the adopted equations and not imposed by clamping or by a narrative assumption.\n\nThe retrograde friction report reports a numerical transition at r
$approx 12.10055$
,
$textkpc$, while the nonlinear-friction node describes a sign change near r
$approx 12.11$
,
$textkpc$. These values are consistent at the level of rounding, but the precise reported value should be tied to a versioned input, equation, and calculation. The existing node also cautions that the friction law becomes anti-damping beyond its sign change; that consequence should be preserved in interpretation rather than described as a generic stable retrograde drag.

5 Discussion: Hawking Evaporation Density Continuity and PID Mapping

Radial density continuity. The project's radial baseline uses a popcorn-type density form consistent with the exchange-current node, with characteristic coherence and cutoff scales and a representative central density on the order of 10^{-29}

,
$textg/cm^3$. A Hawking-associated radial profile should therefore be specified with an explicit mass-loss law (e.g.,
$dotM$
$propto - M^{-2}$), an assumed central mass, and a continuity argument connecting the evaporative outflow to the adopted density profile. Without those steps, the link remains a conceptual bridge rather than a derived result.\n\n**PID mapping.** The working gains are $K_P = 0.92$, $K_I = 0.35$, $K_D = 0.18$. The axial node proposes interpreting these in terms of jet/vacuum coupling, for example K_P
sim
$sqrtP_{textjet}/(M_{textBH}c^2)$, while explicitly marking such relations as untested hypotheses. This addendum recommends keeping that framing: define each gain's empirical role (proportional recovery, integral settling, derivative damping), state the proposed physical correspondence, and identify the measurement that could falsify it.\n\n**Consistency items.** (1) Align the transition radius to a single versioned value — the report's 12.10 kpc versus the nonlinear-friction node's 12.11 kpc — with the exact equation and inputs. (2) Replace the dangling reference to a Master Note §3 with either an actual section or an explicit "hypothesis under active derivation" note. (3) Keep radial and axial equations in separate, clearly labeled blocks to avoid implicitly claiming that the established fit already includes the dipole extension. (4) Note that the two-front retrograde interpretation and Hawking-linked density profile are not yet represented in the current bundle's operational tests, so they should be introduced as proposed extensions with planned verification.

6 Conclusion and Recommended Revisions

The QVEX-NGC6503 framework can accommodate the axial-dipole, retrograde-friction, and Hawking-evaporation ideas as a coherent extension, provided the project preserves the distinction between established parameterization and proposed physical interpretation. The Master Note and the exchange-current node indicate which parts of this addendum are already represented in the repository and which are new. The following revisions would make that distinction durable:\n\n1. Add an explicit section (or clearly marked placeholder) where the Master Note currently expects one, so axial-node references resolve without ambiguity.\n2. Standardize the transition-radius value and show the exact computation, including inputs and rounding, in a reproducible script or notebook.\n3. Tag the two-front retrograde mechanism and the Hawking-density link as proposed extensions with defined tests, rather than presenting them as existing project results.\n4. Document units, boundary conditions, and falsifiable observables for the dipole potential and the gain-mapping hypothesis.\n\nWith these steps, the addendum's concepts can be added to the project in a way that strengthens internal consistency while keeping claims proportionate to current evidence.

7 References

1. A. Gachkar, QVEX_NGC6503_Axial_Constraint_Master_Note.md, Independent Research Note, 2026-10-26 (QVEX-NGC6503-Quantum-Vacuum archive).
2. nodes_analysis/04_Dipole_Axis_Constraint_Node.md, 2026-10-26 (QVEX-NGC6503-Quantum-Vacuum archive).
3. nodes_analysis/02_Nonlinear_Foam_Friction_Node.md, revised 2026-10-29 (QVEX-NGC6503-Quantum-Vacuum archive).
4. nodes_analysis/05_Vacuum_Exchange_Current_Node.md (QVEX-NGC6503-Quantum-Vacuum archive).
5. retrograde_vacuum_friction_report.pdf and retrograde_sign_change.png, user-generated session report for QVEX-NGC6503.